



Effect of kaempferol from *Acori Tatarinowii Rhizoma* (ATR) on grass carp (*Ctenopharyngodon idella*) infected with *Aeromonas hydrophila*

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Abstract

Natural products are effective in the treatment and the prevention of human, animal and plant diseases. Therefore, natural products may also be considered to treat fish diseases. *Acori Tatarinowii Rhizoma* (ATR) is a herbal medicine with anti-inflammatory and antioxidant effects. However, little is known about how its active ingredients exert the beneficial effects. Here, Four effective active ingredients of ATR and their 81 targets were investigated, which affected the anti-inflammatory response. Among them, kaempferol-JUN was identified as a key regulatory module in anti-inflammatory immune responses, and kaempferol interacted with the CiJUN protein and inhibited CiJUN levels. Silencing CiJUN gene in *Ctenopharyngodon idella* kidney (CIK) cells enhanced anti-inflammatory activity and resistance to *Aeromonas hydrophila*, whereas anti-inflammatory activity and resistance were impaired after CiJUN overexpression. The mortality rate of diseased grass carp was reduced after treatment with kaempferol, as were the inflammatory and oxidant effects. Also, grass carp showed enhanced anti-inflammatory and antioxidant effects after feeding with kaempferol. The results provide further insights into the use of kaempferol to prevent and treat fish diseases.

Keywords *Acori Tatarinowii Rhizoma* · Kaempferol · JUN · Anti-inflammatory · Antioxidant · Grass carp

Introduction

Aeromonas hydrophila is a Gram-negative bacterial pathogen that causes serious disease problems (Dong et al. 2021) resulting in economic losses of millions of dollars to freshwater fish farming, worldwide (Yun et al. 2020). The pathogen causes hemorrhagic septicemia in fish by infiltration of the blood stream and release of toxins to induce an excessive

inflammatory response, leading to massive mortalities in freshwater fish (Niu et al. 2023). Inflammation may cause overproduction of ROS, contributing to cell damage in the body (Kadioglu et al. 2015). Therefore, increasing anti-inflammatory activity can induce an antioxidant effect in fish, and this is a means combatting disease caused by *A. hydrophila*.

A. hydrophila was first reported as a pathogen of humans in 1968 (von Graevenitz et al. 1968). In particular, this pathogen has the potential for transmission to humans via food (Dong et al. 2021). Prevention and control of *A. hydrophila* in freshwater fish culture is beneficial not only for the healthy breeding of freshwater fish but also for human health. Antibiotics have been used widely to control diseases caused by *A. hydrophila*, whereas their overuse results in the emergence of resistance, their accumulation in fish tissues, and environmental contamination (Ayoub et al. 2024; Ding and He 2010; Fu et al. 2024). Thus, the failure of antibiotic treatment has led to calls for novel compounds or strategies to combat resistant *A. hydrophila* infections in aquaculture.

Natural products have attracted increasing attention due to their resourcefulness and diversity of biological activity, and are involved in treating and preventing human diseases

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(Melander et al. 2020). Natural products may be developed into products to treat inflammation and oxidation caused by some diseases as alternatives to synthetic compounds (Cheung et al. 2016). Some natural products may possess anti-inflammatory and antioxidant properties in vitro, and may inhibit the production of key inflammation-related cytokines in vivo (Henrotin and Mobasher 2018). The different natural products from the scouring rush (*Equisetum hyemale* L.) have antioxidant and anti-inflammatory activities, displaying inhibition of inflammatory signaling factor in RAW 264.7 macrophages (Jiang et al. 2012). Triptolide, a diterpenoid triepoxide, is used to treat inflammation. Triptolide inhibits NF- κ B-dependent transcription, and suppresses the release of TNF- α and IL-6 in RAW264.7 cells induced by LPS, leading to its anti-inflammatory effect (Ge et al. 2022). Also, natural products affect fish anti-inflammatory and antioxidative responses. Thus, rice bran oil enhances anti-inflammatory and antioxidant properties in zebrafish after copper sulfate treatment (Liu et al. 2023). Reyes-Becerril et al. (2019) report *Chenopodium ambrosioides* L. extract as an interesting additive in functional diets, enhancing the antioxidant and anti-inflammatory effects to improve intestinal health status of *Lutjanus peru*. Moreover, inflammatory responses caused by feeding a high soybean oil diet to large yellow croaker could be alleviated by octanoate supplementation (Zhao et al. 2023).

Acori Tatarinowii Rhizoma (ATR, Shi Chang Pu in Chinese) is the dried rhizome of *Acorus tatarinowii* Schott, a perennial herb of Araceae Juss (Wen et al. 2023). ATR is used to cure a wide range of diseases, including epilepsy, cancers and skin diseases (Wen et al. 2023). Also,

it has anti-inflammatory and antioxidant pharmacological effects. For example, Song et al. (2018) determined that the anti-inflammatory and immune-enhancing effects of ATR are responsible for the ability to treat Alzheimer's disease. Thus, ATR has the potential as a novel natural source of antioxidative agents (Zhang et al. 2015). ATR essential oil has antioxidant and anti-inflammatory properties associated with several groups of compounds, including terpenoids, phenylpropanoids, and aromatic compounds (Wen et al. 2023; Yan et al. 2020). However, it is not known which component plays an anti-inflammatory role. Here, the anti-inflammatory ingredients of ATR were identified and their molecular mechanisms were examined in grass carp. The workflow scheme of this study consisted of four main parts, including identification of the key anti-inflammatory modules, verification of their possible interaction in the modules, and functional analysis of the key target genes and the key active ingredients (Fig. 1).

Materials and methods

Screening of ATR-active ingredients

The MOL2 files and the pharmacokinetic properties of ATR ingredients were downloaded from the TCMSP database, (<https://tcmbspw.com/index.php>). We screened to obtain the active ingredient based on the pharmacological profile of oral bioavailability (OB) $\geq 30\%$ and drug-likeness (DL) ≥ 0.18 .

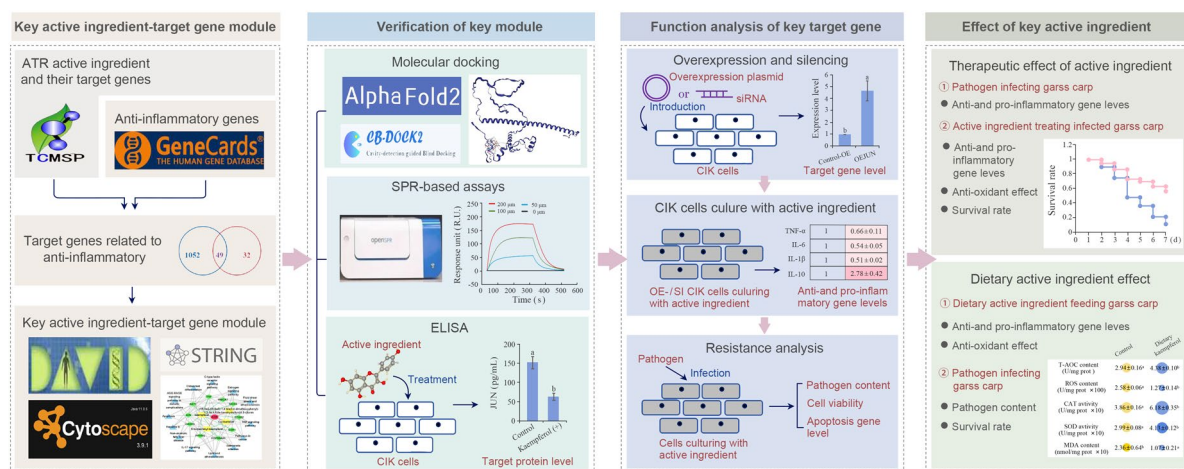


Fig. 1 Workflow scheme. This work was composed of four main parts, including identification of the key active ingredient-target gene module associated with anti-inflammatory activity; verification by

molecular docking, SPR and ELISA; functional analysis of key target genes; and analysis of the anti-inflammatory effect of key active ingredients

Clustering of anti-inflammatory-related genes in ATR

The target genes of the active ingredients were clustered and their gene names were standardized using the TCMSP database and UniProt database, respectively. The anti-inflammatory-related genes from GeneCard database were collected (<https://www.genecards.org>) through searching the key word “anti-inflammatory”. In the search results, the genes that scored higher than the average and that code for proteins were identified as candidates for anti-inflammatory activity. The Veeney 2.1 intersection (<https://bioinfo.gp.cnb.csic.es/tools/venny/>) was used to obtain the potential target genes (i.e., overlapping genes between ATR target genes of active ingredients and anti-inflammatory candidate genes) of ATR for anti-inflammatory effects.

Identification of the hub target genes

The STRING database version 11.5 was used to construct a protein–protein interaction (PPI) network. The target protein sequences were entered, the organism selected as “*Homo sapiens*” and then the search started. The Cytoscape software was used to show the PPI network, and the hub genes were identified by the cytoHubba plugin. Top 10 nodes ranked by degree.

KEGG pathway enrichment analysis

The DAVID software (<https://david.ncifcrf.gov/>) was used to perform the KEGG pathway enrichment analysis. The gene names were submitted to “Gene List” in DAVID, and “*Homo sapiens*” was selected as the background. The pathways with Benjamini < 0.01 were the enriched KEGG pathways.

Targets of the ATR-active ingredient in grass carp

These protein sequences coded by target genes were used as queries to perform multiple database searches against the protein sequences of the *C. idella* genome (assembly HZGC01) (<https://www.ncbi.nlm.nih.gov/genome/?term=Ctenopharyngodon+idella>) by BLASTp searches to acquire the candidate proteins with an E value of 10^{-5} . In each BLAST search, the sequence with the highest score was its target gene in grass carp. Multiple protein sequence alignments were performed with ClustalX1.83.

Docking calculation and surface plasmon resonance assay

AlphaFold2 was used to simulate the three-dimensional (3D) structures of the hub proteins. Molecular docking between the active ingredients and the hub proteins was performed

using CB-DOCK2. The docking structures with the lowest Vina score were visualized.

An open surface plasmon resonance (SPR) instrument (Nicoyalife, Canada) was used to perform SPR analysis. The hub protein as the receptor was prepared by Sangon Biotech, Shanghai. The detailed operation was described by Cui et al. (2024).

CIK cell culture and treatment with kaempferol

The *C. idella* kidney (CIK) cell lines were cultured at 28 °C in M199 containing 10% FCS, 100 µg/mL penicillin, and 100 µg/mL streptomycin (Wu et al. 2019). After digestion by trypsin, cells at a density of 1.5×10^5 cells/cm² were seeded into each well of a 96-well microtiter plate (flat bottom shape). A stock solution of kaempferol (Sigma) was prepared in dimethyl sulfoxide (DMSO) at 100 mmol/L and stored at −20 °C. Then, kaempferol (final concentration 10 µmol/L) was added to each well, and incubated for 24 h. After that, the cells were collected and examined for CiJUN levels using the enzyme-linked immunosorbent assay (ELISA). ELISA kit and assay were performed and provided by HEPENG BIOLOGICAL, Shanghai.

CiJUN overexpression and silencing in CIK cells

The open reading frame (ORF) of *CiJUN* was cloned from grass carp and inserted into the pcDNA3.1 (+) vector to construct the expression plasmid pcDNA3.1-*CiJUN* for the overexpression experiment. The specific siRNA sequence silencing *CiJUN* was developed by DSIR (<http://biodev.extra.ccea.fr/DSIR/DSIR.html>), and the negative control RNA oligo came from the studies of Wu et al. (2016, 2019) (Supplementary Table S2). Introduction of the pcDNA3.1-*CiJUN* plasmid and siRNA sequence into the CIK cells was performed as described by Wu et al. (2016, 2019).

A. hydrophila challenge of CIK cells

The CIK cell lines overexpressing and silencing CiJUN and controls were cultured at 28 °C in M199 containing 10% fetal calf serum (FCS), 100 µg/mL penicillin, and 100 µg/mL streptomycin. After digestion by trypsin, cells at a density of 1.5×10^5 cells/cm² were seeded into each well of a 6-well plate (flat bottom shape). *A. hydrophila* was isolated, identified and stored in our laboratory. CIK cells were treated with *A. hydrophila* (10 µL, 1×10^7 CFU/mL) or PBS (control) for 24 h. After that, the cells were collected to examine their cell viability (Cell Counting Kit-8, Sangon Biotech, Shanghai), and total RNA was extracted to examine the gene expression.

A. hydrophila challenge of grass carp and treatment with kaempferol

Grass carp (length approximately 10 cm) was used to conduct *A. hydrophila* infection experiments in the State Key Laboratory of Developmental Biology of Freshwater Fish. Healthy and similar-sized juvenile grass carp was randomly allocated into nine tanks (1 m × 1 m × 1.5 m), and divided into three groups: the control group, the infection group and the treatment group (3 replicates of 20 fish per group). The rearing conditions and methods were those described by Cui et al. (2024). *A. hydrophila* was isolated, identified, and stored in our laboratory. After 3-day feeding, 100 µL of *A. hydrophila* (1×10^7 CFU/mL) was injected intraperitoneally into the grass carp of the infection and treatment groups, and 100 µL PBS was injected into the control group. 200 µL kaempferol solution in saline (10 mg/mL) was injected into the fish of the treatment group at one day post infection, and maintained in aerated freshwater for seven days. The death of fish was recorded daily, and the cumulative survival rate for each day was calculated to plot the survival curve. The liver samples were harvested, and the blood was collected from the caudal peduncle of the fish with a heparin syringe and placed in nonheparinized tubes. These samples were used for antioxidant activity evaluations and examining gene expression levels.

Feeding grass carp with dietary kaempferol

Healthy and similar-sized juvenile grass carp (weighting approximately 5 g) was randomly allocated into six tanks (1 m × 1 m × 1.5 m). Fish were divided into two groups: the control group and the dietary kaempferol group (three replicates of 20 fish per group). The rearing conditions and methods were the same as above. In the basal diet, vitamin mix and mineral mix were based on the findings of Huang et al. (2023). Control and kaempferol groups were fed the basal diet without and with kaempferol (20 mg/kg), respectively, for a 4-week period (Supplementary Table S1). The feeding methods were those described by Cui et al. (2024). Then, the resistance of the grass carp to *A. hydrophila* was analyzed using the methods described above.

Measurements of oxidation-relevant indicators

Total antioxidant capacity (T-AOC), SOD and CAT activities, and ROS and MDA levels were measured with commercial kits from Nanjing Jiancheng Institute (Nanjing, China) according to the manufacturer's instructions.

Isolation of total RNA and RT-qPCR analysis

Total RNA from all samples was isolated using TRIzol reagent (TaKaRa, Dalian, China), and the first-strand cDNA was synthesized using a PrimeScript™ RT-PCR kit (TaKaRa, Dalian, China). All operations were performed according to the manufacturer's instructions.

RT-qPCR was performed with a SYBR Premix Ex Taq™ II Kit (TaKaRa) on an ABI7500 platform according to the manufacturer's instructions. Primer information is shown in Supplementary Table S2. The relative changes for each gene were analyzed by the $2^{-\Delta\Delta C_t}$ method (Jiang et al. 2019).

Results

ATR-active ingredient

A total of 105 ATR ingredients were found from the TCMSP database. Only four active ingredients met the screening criteria of OB ≥ 30% and DL ≥ 0.18. These were 8-isopentenyl-kaempferol, cycloartenol, kaempferol, and (1R,3aS,4R,6aS)-1,4-bis(3,4-dimethoxyphenyl)-1,3,3a,4,6,6a-hexahydrofuro[4,3-c] furan (Supplementary Fig. S1).

Anti-inflammatory-related genes in ATR

Four active ingredients of ATR acted on 81 target genes. We selected “anti-inflammatory” as a key word, and searched for it in the GeneCard databases. A total of 5276 searched entries were obtained, including 3713 protein-coding genes. Among these, 1101 genes were obtained as the anti-inflammatory-related genes with relevance scores greater than the average score. ATR target genes of active ingredients and anti-inflammatory genes were intersected to obtain 49 potential targets as the anti-inflammatory-related genes in ATR (Fig. 2A, Supplementary Table S3). Then, we obtained the sequences of these proteins from the GeneCard database. Thus, there were two sequences for both estrogen receptor and heat shock protein HSP90, such as ESR1, ESR2, HSP90AA1, and HSP90AB1, respectively.

Key anti-inflammatory targets in ATR

The PPI network contains 51 nodes (Fig. 2B). After importing the PPI network into Cytoscape, based on the node scores, the top 10 nodes showed that the deeper the color of the gene, the higher the ranking (Fig. 2C). These genes included AKT1, JUN, TNF, and ESR1.

These 10 hub genes were involved in 13 KEGG pathways (Benjamini < 0.01), representing key pathways closely

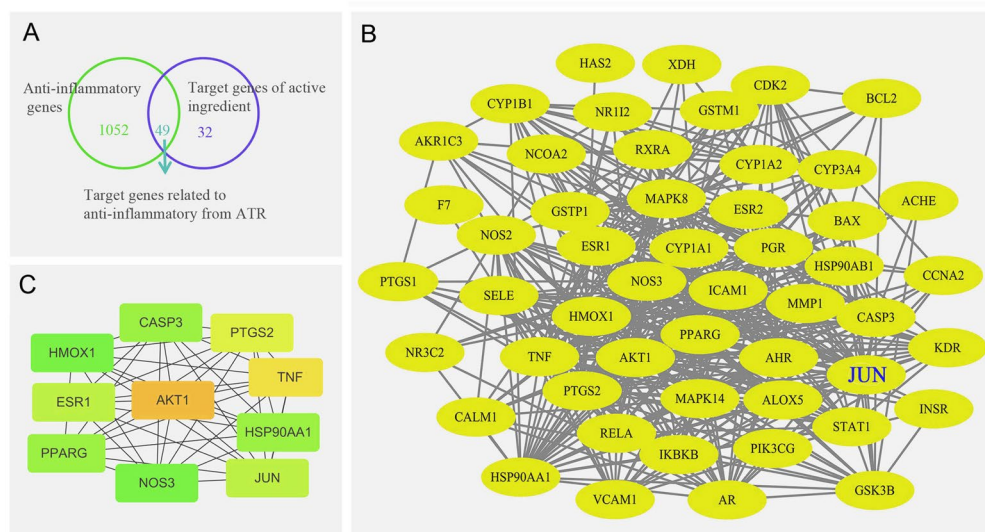


Fig. 2 Potential target genes and PPI network map of ATR therapy for anti-inflammation. **A** Venny results of potential target genes of ATR therapy for anti-inflammation; **B** the PPI network of potential target genes using the STRING database; **C** the hub target genes of

the PPI network. The hub genes were identified using the cytoHubba plugin of Cytoscape. The deeper the color of the gene in network graph is, the higher the ranking is

associated with the anti-inflammatory activity of ATR. These pathways included IL-17 signaling pathway, *Salmonella* infection, lipid and atherosclerosis, and TNF signaling pathway (Table 1).

Kaempferol-CiJUN gene module

To explain the regulatory relationship of active ingredients, hub targets and key pathways, the active ingredient-hub target-key pathway network was constructed. As shown in

Fig. 3A, only kaempferol could interact with all hub proteins. In addition, of these 10 hub proteins, only JUN was involved in all the enrichment pathways. Overall, kaempferol-JUN was the key regulated module.

To determine whether kaempferol acts on *C. idella* JUN (CiJUN), we analyzed the homologous relationship between human JUN and CiJUN. A CiJUN (XP_051731871.1) was identified as the human JUN homologous protein with an E value of 2×10^{-174} , and the highest identities (75.30%). The multiple alignment between them showed that their

Table 1 Pathways containing the key genes

Pathway	Gene	Benjamini
Lipid and atherosclerosis	<i>HSP90AA1, JUN, NOS3, CASP3, AKT1, PPARG, TNF</i>	3.4×10^{-6}
Pathways in cancer	<i>HSP90AA1, JUN, CASP3, HMOX1, AKT1, PPARG, PTGS2, ESR1</i>	7.5×10^{-6}
Fluid shear stress and atherosclerosis	<i>HSP90AA1, JUN, NOS3, HMOX1, AKT1, TNF</i>	7.5×10^{-6}
IL-17 signaling pathway	<i>HSP90AA1, JUN, CASP3, PTGS2, TNF</i>	7.3×10^{-5}
AGE-RAGE signaling pathway in diabetic complications	<i>JUN, NOS3, CASP3, AKT1, TNF</i>	7.5×10^{-5}
TNF signaling pathway	<i>JUN, CASP3, AKT1, PTGS2, TNF</i>	1.1×10^{-4}
Estrogen signaling pathway	<i>HSP90AA1, JUN, NOS3, AKT1, ESR1</i>	1.9×10^{-4}
Non-alcoholic fatty liver disease	<i>JUN, CASP3, AKT1, PPARG, TNF</i>	2.7×10^{-4}
<i>Salmonella</i> infection	<i>HSP90AA1, JUN, CASP3, AKT1, TNF</i>	1.5×10^{-3}
C-type lectin receptor signaling pathway	<i>JUN, AKT1, PTGS2, TNF</i>	2.4×10^{-3}
Osteoclast differentiation	<i>JUN, AKT1, PPARG, TNF</i>	4.1×10^{-3}
Apoptosis	<i>JUN, CASP3, AKT1, TNF</i>	4.5×10^{-3}
Hepatitis B	<i>JUN, CASP3, AKT1, TNF</i>	6.9×10^{-3}

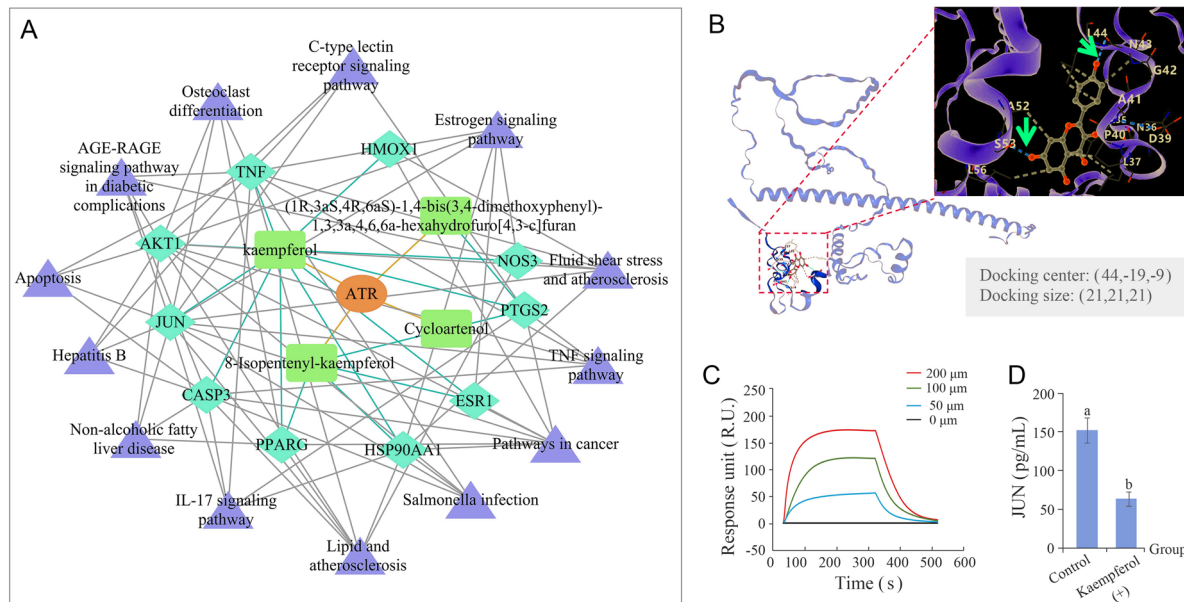


Fig. 3 Interactions between the kaempferol and the CiJUN protein. **A** ATR-active ingredient-hub target-key KEGG pathway network. Triangles represent the enrichment pathways; diamonds represent the hub proteins; rectangles represent the active ingredients; ellipse represents ATR; **B** the binding pattern between kaempferol and the CiJUN protein in the molecular docking model. Arrows represent

hydrogen bonds; **C** SPR assay showing interactions between the kaempferol and the CiJUN protein; **D** ELISA results showing the CiJUN protein level in CIK cells after kaempferol treatment. All the experiments were repeated at least three times. Error bar, mean $\pm$ SD. Letters indicate significant differences among samples ($P < 0.05$)

sequences were conserved (Supplementary Fig. S2). Also, the human JUN and CiJUN structures were predicted using AlphaFold2. Their 3D-structures were almost identical, especially an alpha helix located in the N-terminus of both human JUN and CiJUN, which may be associated with the formation of dimers to affect function (Supplementary Fig. S3).

The result of the molecular docking with the lower Vina scores between kaempferol and CiJUN, suggesting that the binding is more stable, was shown in Fig. 3B. Molecular docking center and docking size were (44, -19, -9) and (21, 21, 21), respectively. There were two hydrogen bonds in the molecular docking complex: ALA52 and ASN43 of the CiJUN protein formed with kaempferol. The equilibrium dissociation constant (KD) for kaempferol was 4.86×10^{-4} using SPR, demonstrating that kaempferol can act on CiJUN (Fig. 3C). In addition, after CIK cells were cultured together with kaempferol, the CiJUN protein level was significantly decreased (Fig. 3D). These results suggested that kaempferol may inhibit the CiJUN protein.

CiJUN affects disease resistance in *C. idella*

To explore the effect of CiJUN, *CiJUN* was overexpressed and silenced in CIK cells, respectively. The expression level of *CiJUN* was upregulated 4.67-fold in the CIK cells

overexpressing *CiJUN* compared to that of the control (Fig. 4A). *CiJUN* expression was inhibited in the CIK cells silenced for *CiJUN* using siRNA methods (Fig. 4B). The inflammatory response in fish involves antagonism of some cytokines. *TNF- α* , *IL-6*, and *IL-1 β* increase inflammation, while *IL-10* decreases inflammation. After overexpression of *CiJUN* in CIK cells, *TNF- α* , *IL-6*, and *IL-1 β* expression were promoted, while *IL-10* was inhibited (Fig. 4C). Conversely, *CiJUN* was silenced in CIK cells, leading to a decrease in *TNF- α* , *IL-6*, and *IL-1 β* levels and an increase in *IL-10* levels (Fig. 4D). These results suggest that CiJUN may promote inflammatory activity in CIK cells.

CIK cells overexpressing or silencing for *CiJUN* were challenged with *A. hydrophila*. Cell viability was determined at approximately 24 h post infection. Viability of CIK cells overexpressing *CiJUN* was significantly decreased, at only 53.44% of the cell viability of the control (Fig. 4E). Caspases (CAP) and Bax are involved in the apoptotic responses of cells to different stimuli. The transcript levels of *CAP-3*, *-8*, and *Bax* were increased in CIK cells overexpressing CiJUN compared to the control cells after infection with *A. hydrophila* (Fig. 4F). The *16S rRNA* levels, as measured by RT-qPCR, was used to show the abundance of *A. hydrophila* in CIK cells. After infection, *A. hydrophila* abundance was significantly high in the CIK cells overexpressing *CiJUN* compared to control cells (Fig. 4G). Instead, the viability of CIK cells with *CiJUN*

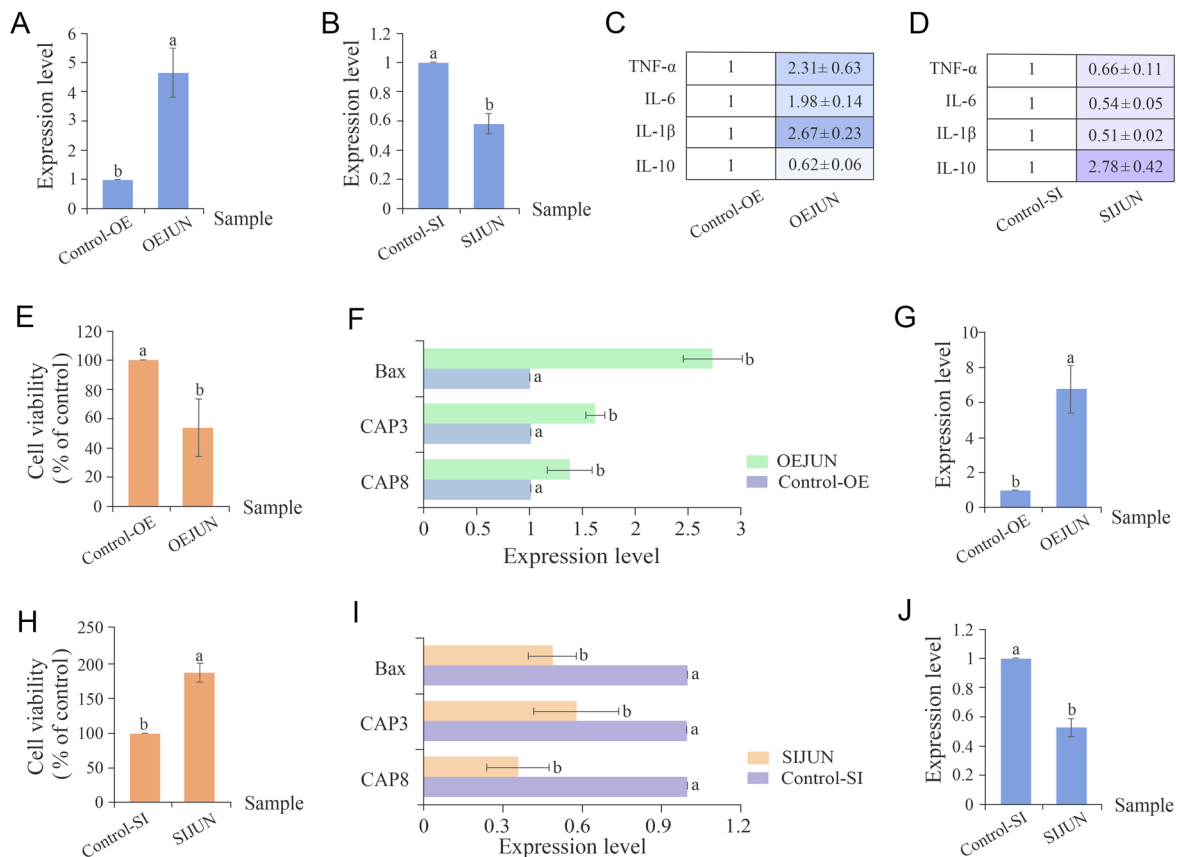


Fig. 4 *CiJUN* affects the anti-inflammatory effects and resistance of CIK cells to *A. hydrophila*. **A**, **B** Relative expression levels of *CiJUN* in CIK cells overexpressing (**A**) and silencing (**B**) *CiJUN*; **C**, **D** expression levels of pro- (*TNF- α* , *IL-6*, *IL-1 β*) and anti-inflammatory (*IL-10*) genes in CIK cells overexpressing (**C**) and silencing (**D**) *CiJUN*. The deeper the color is, the higher the expression level is; **E** cell viability was evaluated in *A. hydrophila*-induced CIK cells overexpressing *CiJUN*; **F** Apoptosis-related gene (*CAP-3*, *-8* and *Bax*) levels in CIK cells overexpressing *CiJUN* after infection with *A. hydrophila*. **G** Accumulation of the transcript of *A. hydrophila* 16S rRNA showing *A. hydrophila* content in CIK cells overexpress-

ing *CiJUN* after infection with *A. hydrophila*. **H** Cell viability was evaluated in *A. hydrophila*-induced CIK cells with *CiJUN* silencing; **I** apoptosis-related gene levels in CIK cells with *CiJUN* silencing after infection with *A. hydrophila*. **J** Accumulation of the transcript of *A. hydrophila* 16S rRNA in CIK cells with *CiJUN* silencing after infection with *A. hydrophila*. All the experiments were repeated at least three times. Error bar, mean $\pm$ SD. Letters indicate significant differences among samples ($P < 0.05$). Grass carp β -actin was used as a reference control gene for RT-qPCR analysis. OE stands for overexpression and SI stands for silencing

silencing was significantly increased after 24 h of infection with *A. hydrophila* (Fig. 4H). The transcript levels of the *CAP-3*, *-8* and *Bax* genes were lower than those in the control (Fig. 4I). The abundance of *A. hydrophila* decreased significantly in CIK cells with *CiJUN* silencing compared to control cells (Fig. 4J). These results suggested that *CiJUN* negatively affects *C. idella* resistance to *A. hydrophila* by promoting inflammatory-related gene expression.

Kaempferol positively contributes to anti-inflammation

To clarify the effect of kaempferol during the fish anti-inflammatory process, the grass carp was infected with

A. hydrophila for one day to construct a model that produces an inflammatory response. Then, they were treated with kaempferol, and their anti-inflammatory ability was examined at seven days post treatment (Fig. 5A). After infection with *A. hydrophila*, the expression levels of *TNF- α* , *IL-6*, and *IL-1 β* were significantly up-regulated in the liver and blood samples of the infection group compared with those in the control group (Fig. 5B). This result suggested that grass carp undergoes an inflammatory reaction after infection with *A. hydrophila*. The *TNF- α* , *IL-6*, and *IL-1 β* levels had a significant relationship with kaempferol treatment (Fig. 5C). The transcript levels of *TNF- α* , *IL-6*, and *IL-1 β* were significantly down-regulated in the treatment groups at seven days after treatment

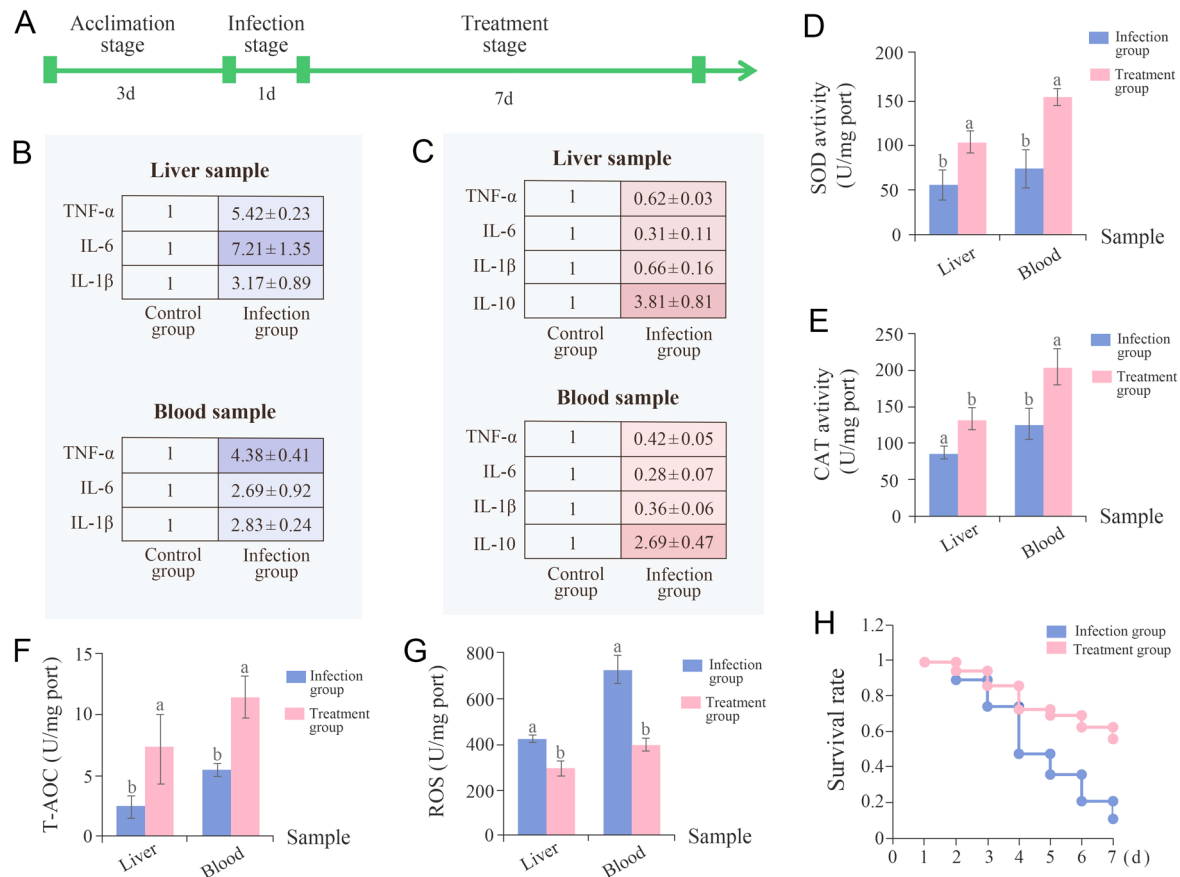


Fig. 5 Kaempferol positively contributes to anti-inflammatory and antioxidant after *A. hydrophila* infection. **A** Schematic showing the experimental procedure for the kaempferol treatment of grass carp infected with *A. hydrophila*; **B** expression levels of *TNF- α* , *IL-6*, and *IL-1 β* genes in the liver and blood samples of grass carp infected with *A. hydrophila*. The deeper the color is, the higher the expression level is; **C** expression levels of *TNF- α* , *IL-6*, *IL-1 β* and *IL-10* genes in the liver and blood samples of the treatment group of grass carp

at 7 days. The deeper the color is, the higher the expression level is; **D–G** SOD activity (**D**), CAT activity (**E**), T-AOC (**F**) and ROS level (**G**) in the treatment group at seven days; **H** Cumulative survival rate of grass carp treated with kaempferol after infection with *A. hydrophila*. All the experiments were repeated at least three times. Error bar, mean $\pm$ SD. Letters indicate significant differences among samples ($P < 0.05$). Grass carp *β -actin* was used as a reference control gene for RT-qPCR analysis

with kaempferol, approximately 0.62-fold, 0.31-fold, and 0.66-fold in liver samples and 0.42-fold, 0.28-fold, and 0.36-fold in blood samples, respectively, and the transcript level of *IL-10* was significantly up-regulated.

Moreover, the levels of T-AOC and the activities of SOD and CAT were also affected by kaempferol. Compared with the infection group, they were all elevated in liver and blood samples of the treatment group, whereas ROS content was decreased in the treatment group (Fig. 5D–G). At seven days post treatment with kaempferol, the survival rate of the treatment group (56.7%) is high compared to that of the infection group (11.7%) (Fig. 5H).

Dietary kaempferol enhances disease resistance in *C. idella*

Grass carp was fed with kaempferol. After four weeks, *A. hydrophila* was used to infect the grass carp. Indeed, the resistance of the grass carp was examined (Fig. 6A). The grass carp in the dietary kaempferol group showed decreased MDA contents, increased levels of T-AOC and the activities of the SOD and CAT, and decreased ROS levels (Fig. 6B).

In the dietary kaempferol group, the *TNF- α* , *IL-6*, and *IL-1 β* expressions were significantly down-regulated in both liver and blood samples, whereas *IL-10* was up-regulated (Fig. 6C, D). The grass carp in the dietary kaempferol group

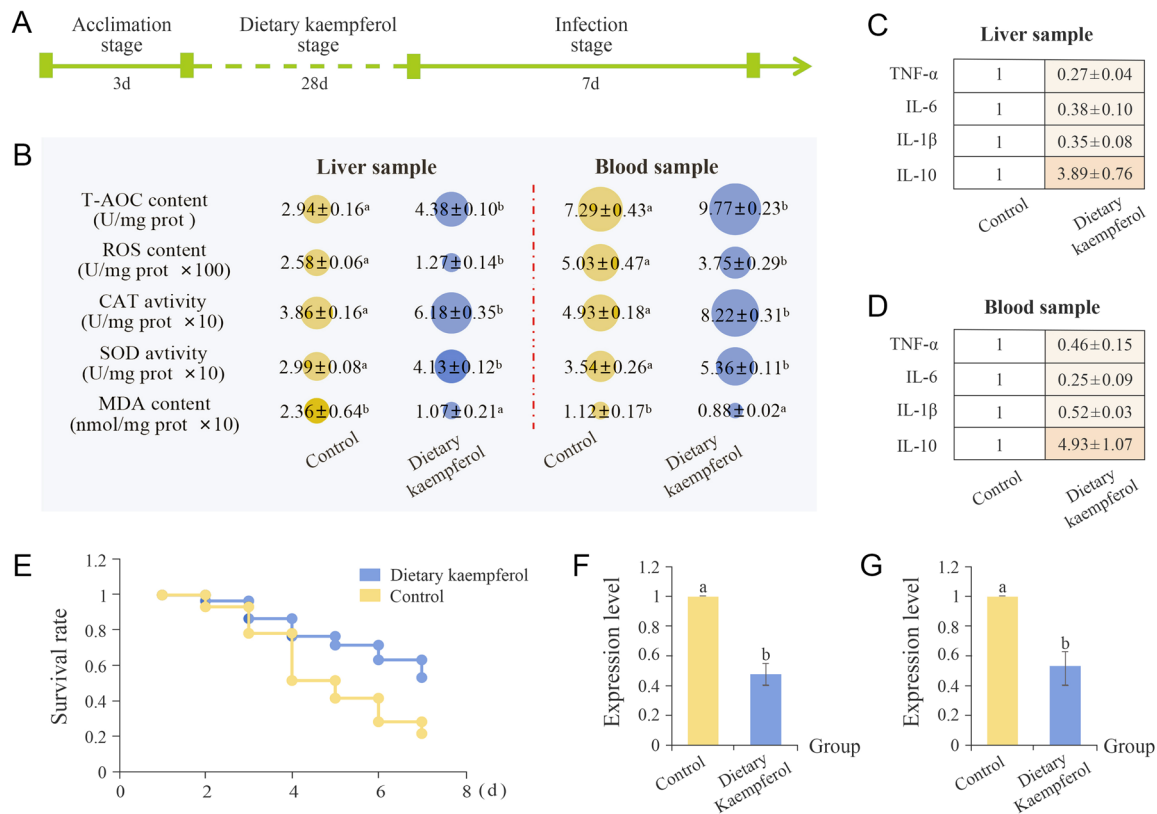


Fig. 6 Dietary kaempferol enhances the anti-inflammatory, antioxidant and resistance of grass carp to *A. hydrophila*. **A** Schematic showing the experimental procedure for the grass carp infected with *A. hydrophila* after feeding with a basal diet supplemented with kaempferol; **B** T-AOC, ROS level, SOD activity, CAT activity and MDA content in the liver and blood samples of the grass carp fed with the basal diet supplemented with kaempferol; **C**, **D** Expression levels of pro- (*TNF-α*, *IL-6*, *IL-1β*) and anti-inflammatory (*IL-10*) genes in the liver and blood samples of the grass carp fed a basal

diet supplemented with kaempferol. **E** Cumulative survival rate of grass carp infected with *A. hydrophila* after feeding with the basal diet supplemented with kaempferol; **F**, **G** Accumulation of the transcript of *A. hydrophila* 16S rRNA in the liver (**F**) and blood (**G**) samples of grass carp infected with *A. hydrophila* after feeding with the basal diet supplemented with kaempferol. All the experiments were repeated at least three times. Error bar, mean ± SD. Letters indicate significant differences among samples ($P < 0.05$). Grass carp β -actin was used as a reference control gene for RT-qPCR analysis

had a higher survival rate than the control group after a 7-day challenge with *A. hydrophila* (Fig. 6E). In addition, *A. hydrophila* abundance in the control group was significantly up-regulated compared to that in the dietary kaempferol group (Fig. 6F, G).

Discussion

A. hydrophila triggers inflammatory responses and oxidative stress in fish

A. hydrophila is a freshwater, facultatively anaerobic pathogen that causes disease in fish and mammals. Rasmussen-Ivey et al. (2016) explained that *A. hydrophila* causes disease by secreting multifactorial virulence factors, and by mediating of virulence factor expression through quorum sensing.

Once this pathogen infiltrates the bloodstream and releases virulence factors in fish, an excessive inflammatory response and oxidative stress will be induced, leading to gastroenteritis and septicemia (Niu et al. 2023). Thus, improvement of anti-inflammatory activity and reduction of oxidation in fish are a means of managing disease caused by *A. hydrophila*.

Fish liver is an important immune organ that plays a defensive role during infection with various pathogens (Dezfuli et al. 2023). For example, the levels of ITGβ1, C3, C5, and MRC1 from *Cynoglossus semilaevis*, were altered in the liver after infection with *Vibrio anguillarum*, affecting disease resistance (Qi et al. 2021). In the liver of Indian catfish, *Clarias magur*, the *Toll-like receptor 2* gene expression was significantly increased at 8 h and 24 h post infection to *A. hydrophila* (Muduli et al. 2021). In liver tissue, inflammation and oxidants are the most important biological responses induced by *A. hydrophila*. When *A. hydrophila*

invades gibel carp (*Carassius auratus gibelio*), the balance between ROS and antioxidant enzymes is disturbed, and inflammation-related signaling pathways, such as Toll-like receptors (TLRs), are activated (Gu et al. 2022). Meanwhile, infection of *C. auratus* by *A. hydrophila* leads to decreased T-AOC, increased ROS levels and activation of TLR signaling pathway in blood (Xu et al. 2020).

Kaempferol as a key active ingredient in ATR improves the resistance of grass carp against *A. hydrophila*

To enhance anti-inflammatory activity and reduce oxidant effect after infection with *A. hydrophila*, several natural substances have been used to control redox imbalance in fish. For example, dietary supplementation of grass carp with 8% caffeine for one month increased survival rate after infection with *A. hydrophila*, reduced microbial load and prevented lipid and protein damage in the liver (Baldissera et al. 2019). Also, ATR is involved in anti-inflammatory and antioxidant effects. Terpenoids, phenylpropanoids, and aromatic compounds from ATR essential oil have anti-inflammatory and antioxidant effects (Wen et al. 2023; Yan et al. 2020).

Kaempferol is a polyphenolic compound among the active ingredients in ATR, and it is involved in anti-inflammatory and antioxidant processes (Atak et al. 2024). Furthermore, Kadioglu et al. (2015) found that inflammation could cause ROS to become overproduced, contributing to cell damage. In RAW264.7 cells, kaempferol may exert protective effects against LPS-induced sepsis by reducing an inflammatory response (Zhu et al. 2022). Inflammatory response to *Helicobacter pylori* infection was also reduced by kaempferol involving the reduction of proinflammatory cytokine expression (Yeon et al. 2019). Thus, kaempferol may reduce oxidative stress while reducing the inflammatory response. As a novel and promising therapeutic agent, kaempferol may reduce the inflammatory and oxidative stress caused by spinal cord injury in in vivo and in vitro models (Liu et al. 2021). Meanwhile, this study illustrates that kaempferol enhances the anti-inflammatory effect in fish. Inflammation and oxidative stress caused by *A. hydrophila* in grass carp were inhibited after treatment with kaempferol, with upregulation of anti-inflammatory-related genes, downregulation of proinflammatory-related gene, decreased levels of MDA and ROS, and increased levels of T-AOC and the activities of SOD and CAT (Fig. 5). Meanwhile, grass carp fed with kaempferol showed a decrease in ROS levels, an increase in anti-inflammatory activity and an increase in survival rates after infection with *A. hydrophila* (Fig. 6). These results suggest that kaempferol may reduce oxidative stress and inflammation in grass carp to enhance their resistance to *A. hydrophila*.

Kaempferol promotes an anti-inflammatory response by inhibiting the CiJUN protein

Previous studies have shown that the anti-inflammatory effect of kaempferol is mediated by the inhibition of AP-1 transcription factor (Chen et al. 2004; Crespo et al. 2008; Yeon et al. 2019). JUN protein is an essential component of the AP-1 transcription factor (Karin et al. 1997). In this study, the kaempferol-JUN module was identified as the key anti-inflammatory module from ATR. The results of the molecular docking, SPR-based assays, and ELISA showed that the kaempferol inhibited and decreased CiJUN levels in grass carp (Fig. 3). Thus, the anti-inflammatory effects of kaempferol in grass carp may be due to its inhibition of the CiJUN protein.

JUN, as a component of the AP-1 transcription factor, binds to target inflammatory genes, regulating their expression. For example, in vertebrates, Jun D could control levels of the proinflammatory cytokine IL-1 β levels but not IL-6 and TNF- α in the brain tissue of mice (Diaz-Cañestro et al. 2019). Meanwhile, the production of the proinflammatory cytokines CgIL17-4 and CgIL17-5 in *Crassostrea gigas* was promoted by AP-1 that contains a Jun domain (Wang et al. 2020). Thus, JUN protein promotes the inflammatory response. Once JUN is inhibited, an anti-inflammatory response will occur. Cereblon degrades JUN to attenuate AP-1 transcriptional activity and reduce several proinflammatory cytokine levels, leading to inhibition of the inflammatory response (Yang et al. 2018). TIRAP transactivates cJUN to promote an inflammatory response to AP1. As the TIRAP-c-Jun interaction is inhibited, AP1-mediated inflammatory responses are also inhibited (Srivastava et al. 2019). In CIK cells, *CiJUN* silencing led to decreased proinflammatory cytokines and increased anti-inflammatory cytokines, whereas the opposite result was observed after *CiJUN* overexpression (Fig. 4). These results suggest that inhibition of CiJUN in grass carp promotes an anti-inflammatory response.

Conclusion

Our results explain how kaempferol confers anti-inflammatory and antioxidant effects in grass carp after interaction with *A. hydrophila* (Fig. 7). Kaempferol interacts with the CiJUN protein and inhibits CiJUN level in grass carp. Once CiJUN protein is inhibited, proinflammatory gene levels are down-regulated, anti-inflammatory gene levels are up-regulated, and antioxidant capacity is enhanced, leading to reduced inflammation and oxidative stress and increased survival rate of grass carp after infection with *A. hydrophila*. These results explain the anti-inflammatory function of

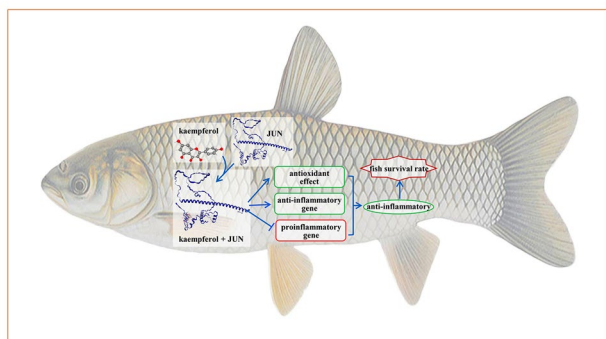


Fig. 7 Model of the kaempferol-CiJUN module involved in the anti-inflammation, antioxidant and resistance of grass carp to *A. hydrophila*

kaempferol and the possibility of using kaempferol to manage fish diseases.

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Author contributions JC and NJ conceptualized and designed the experiment. QF conducted network pharmacology analysis. JC performed the biological experiments. RXZ and ZHZ provided technical assistance. JC and ZHZ analyzed the data and wrote the manuscript. JC, NJ, RXZ, HYB and ZQY corrected the manuscript. All the authors approved the final manuscript.

Data availability Data will be made available on request.

Declarations

Conflict of interest The authors declare that there are no conflicts of interest.

Animal and human rights statement This article does not contain any experiments with human participants conducted by the authors. All the experiments were conducted following standard procedures.

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